

Muhammad Ammaz

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PROFESSIONAL SUMMARY

Full-Stack Software Engineer with hands-on production experience building scalable web applications and AI-powered systems. Proficient in the MERN Stack (MongoDB, Express.js, React, Node.js), REST API design, microservices architecture, and cloud-ready deployment with Docker. Additional expertise in LLM integration, RAG pipelines, and agentic workflows. Proven ability to deliver end-to-end solutions from data ingestion to user-facing interfaces in fast-paced environments.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python
Frontend: React.js, Next.js, Tailwind CSS
Backend: Node.js, Express.js, REST APIs, JWT Authentication
Databases: MongoDB, Microsoft SQL Server, Redis
DevOps & Tools: Docker, Git, Postman, SonarQube
AI & Integrations: LangChain, LangGraph, OpenAI APIs, ChromaDB, RAG Pipelines, Sentiment Analysis
Architecture & Practices: Microservices, Event-Driven Architecture, Agile/SDLC, OOP, Unit Testing (Jest, Mocha)

WORK EXPERIENCE

Full-Stack Software Engineer

January 2026 – Present

GCS Pvt Ltd

Karachi, Pakistan

- Architected and deployed a production-grade Crime Intelligence Platform using LangGraph, LangChain, and OpenAI APIs with a full MERN stack backend, enabling law enforcement agencies to extract insights from large-scale FIR records.
- Built an advanced RAG pipeline with query rewriting and agent-driven dynamic retrieval, improving intelligence summary accuracy over naive RAG approaches.
- Designed multi-agent workflows using LangGraph where specialized agents handle evidence retrieval, pattern analysis, hotspot analysis, and report generation, coordinated through a supervisor agent graph.
- Developed a data ingestion portal with secure validation, CRUD operations, and ChromaDB integration for semantic retrieval of structured and unstructured records.
- Implemented microservices and event-driven architecture with Redis caching to reduce vector search overhead and improve response latency under high concurrency.
- Containerized the full stack with Docker and deployed on-premise with secure port-sharing, meeting strict client data residency requirements.
- Built a Social Media Listening Agent — autonomous agents collecting posts and engagement data across platforms, with NLP-based sentiment analysis and an automated contextual reply module.
- Developed HireFlow, a full-stack AI-powered HR automation platform: resume screening engine, candidate ranking system, AI interview module, and email automation — built with Node.js, Express.js, React.js, and ChromaDB.

MERN Stack Intern

October 2025 – December 2025

10Pearls

Karachi, Pakistan

- Built a production-grade full-stack Notes application with user authentication and full CRUD using the MERN stack.
- Achieved comprehensive test coverage with 18 backend and 6 frontend unit tests using Jest.
- Integrated SonarQube for static code analysis and resolved all flagged code quality issues.
- Implemented secure RESTful JWT authentication with stateless session management and role-based access control.

MERN Stack Intern

August 2025 – September 2025

RealTech Solutions

Karachi, Pakistan

- Updated and maintained modules in a large-scale ERP system built on Node.js and Express.js.
- Authored complex SQL queries against Microsoft SQL Server for management reporting dashboards.
- Designed dashboard interfaces for sales performance, inventory, and operational KPI visualization.
- Optimized API calls and queries, achieving a 20% reduction in page load times.

PROJECTS

Chest Guard: Medical Image Analysis System | *React, Node.js, Express, MongoDB, Python, VGG16* Final Year Project

- Architected a full-stack web app to analyze 4,500+ chest X-rays, detecting Tuberculosis and Pneumonia via a trained VGG16 deep learning model.
- Built Node.js/Express APIs to receive images, forward to the Python model, and return structured predictions; persisted results in MongoDB for physician review.
- Designed a responsive React + Tailwind CSS frontend for image uploads and diagnostic display.

E-Commerce Marketplace | *Next.js, Sanity CMS, Clerk Auth, Stripe* 72-Hour Hackathon

- Delivered a complete online marketplace with product pages, shopping cart, and checkout within 72 hours.
- Integrated Sanity CMS for headless product management, Clerk for authentication, and Stripe for secure payment processing.
- Built an admin dashboard for real-time order viewing and fulfillment status tracking.

EDUCATION

Sir Syed University of Engineering and Technology

Bachelor of Science in Software Engineering (BSSE)

Karachi, Pakistan

Graduated: 2025